

Clustering Documents along Multiple Dimensions

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What is a Clustering Dimension?

- a dimension along which a dataset can be naturally clustered
 - E.g., **Movie reviews** can be clustered by
 - **genre** (action, romantic, documentary, ...)
 - **sentiment** (positive, negative, ...)
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clustering dimensions

- Many natural language documents can be naturally clustered along **multiple** dimensions

Task

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Multi-Clustering

Why bother?

- Exploratory data analysis
 - useful for someone who doesn't know how the data can be clustered

Multi-Clustering

- Can we perform multi-clustering using traditional clustering algorithms?

Multi-Clustering

- Can we perform multi-clustering using traditional clustering algorithms?
 - Perhaps no ...
 - Typically only one clustering is produced

Multi-Clustering

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 - design different similarity functions or objective functions so that multiple meaningful clusterings can be produced

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Defeats the purpose of exploratory data analysis

Problem Formulation

Input: X_n : a set of n documents $\{x_1, x_2, \dots, x_n\}$

z : number of clusterings to be produced

k : z numbers (# clusters to be produced for each clustering)

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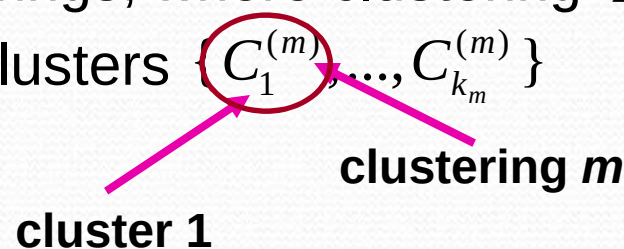
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cluster 1 clustering m

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What criteria should it use to determine whether one multi-clustering solution is better than the other?

Three criteria for determining whether solution K_1 is better than solution K_2

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- **Quality:** K_1 is better than K_2 if K_1 has a higher quality than K_2 according to a **clustering quality preference function**
 - E.g., log likelihood (Dempster et al., 1977)

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Goal: to design an objective function based on these three criteria such that optimizing the function is equivalent to favoring solutions that better satisfy these criteria

Our Approach to Multi-Clustering

- Propose the **Probabilistic Multi-Clustering (PMC)** model
 - a **generative, unsupervised, knowledge-lean** approach
 - assumes a **single feature space** composed of **unigrams**
 - employs an objective function based on the three multi-clustering criteria
 - extends the **Gaussian Mixture Model (GMM)** to produce multiple clusterings

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- can be used to generate a **single** clustering of a data sample

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$$p(x | \Theta) = \sum_{j=1}^k p(C_j | \gamma) p(x | C_j, \mu_j, \Sigma_j)$$

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The diagram illustrates the Gaussian Mixture Model equation with several annotations:

- feature vector**: Points to x in $p(x | \Theta)$.
- number of clusters**: Points to k in the summation index.
- set of model parameters**: Points to Θ in $p(x | \Theta)$.
- mixture weights**: Points to $p(C_j | \gamma)$ in the summation.
- component Gaussian densities**: Points to $p(x | C_j, \mu_j, \Sigma_j)$ in the summation.
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- **Parameter estimation**

- Θ is composed of $3k$ parameters; 3 parameters per component
 - k components; 3 parameters per component (μ_j , Σ_j , and γ_j)

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 - k components; 3 parameters per component (μ_j , Σ_j , and γ_j)
- Θ can be computed by maximizing log likelihood of data

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- **Clustering**

- the learned Θ can be used to induce a soft document clustering
- a hard clustering can be derived from the soft clustering by assigning each point x to its most likely cluster

Multi-Clustering GMM (MGMM)

- Extends the GMM to produce multiple clusterings

A naïve MGMM for generating z clusterings

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3 parameters per cluster
 k_m clusters per clustering
 z clusterings in total

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Favor models that produce dissimilar clusterings

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Favor models that produce dissimilar clusterings

Favor models less prone to overfitting

Implementing Distinctivity Constraints

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A cluster location is defined by the mean of a Gaussian.

- Prefer models with small similarity function values
 - Small values cluster locations are dissimilar
 clusterings in the solution are dissimilar

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Small similarity values better models in term of distinctivity

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Smaller aggregate regularization function values
more parsimonious models

$$\left\{ \begin{array}{l} \eta_{\mu}(\Theta) = \sum_{m,j} \left\| \mu_j^{(m)} \right\|^2 \\ \eta_{\sigma}(\Theta) = \sum_{m,j} \left\| \sigma_j^{(m)} \right\|^2 \\ \eta_{\gamma}(\Theta) = \sum_{m,j} \left\| \gamma_j^{(m)} \right\|^2 \end{array} \right.$$

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Smaller $V(\Theta)$ better models
Better models have larger p_{Θ}

Modifying the Objective Function

1. Incorporate prior p_{Θ} into the objective function used by the naïve MGMM
2. Seek a MAP estimate of the resulting objective function

Parameter Estimation

- employ BFGS in a 5-step multi-stage estimation algorithm

Evaluation

Goal:

Determine the effectiveness of PMC in producing multiple clusterings along the important dimensions of a text dataset

Five Text Datasets

- Each has multiple **2-way** clustering dimensions

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Dataset	Description	Dimensions
BOOK	2000 book reviews	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective)
DVD	2000 DVD reviews	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective)
MIX	4000 reviews from BOOK and DVD	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective) Topic (Book vs. DVD)
MAN	1000 blog posts	Gender (Male vs. Female) Age (Old vs. Young)
POL	2000 political articles	Pol. Affiliation (Dem. vs. Republican) Policy (Foreign vs. Domestic)

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Evaluation Measure

- Accuracy
 - fraction of documents correctly labeled
 - computed for each clustering dimension

Baseline Multi-Clustering Algorithms

- Three baselines
 - **Meta clustering** (Caruana et al., 2006)
 - unsupervised
 - **GMM-EM**
 - unsupervised
 - **Davidson & Qi's** (2007) system
 - semi-supervised

Baseline 1

- Meta clustering (Caruana et al., 2006)
 - For each clustering dimension,
 - run 2-means 100 times, each time with a random selection of seeds and a random weighting of features
 - report best accuracy achieved

Baseline Results: Meta Clustering

	BOOK		MIX			MAN		POL	
	Sent.	Subj.	Topic	Sent.	Subj.	Gend.	Age	Affili.	Policy
Meta clustering	50.8	51.2	50.2	50.2	58.6	51.2	53.6	59.4	58.8

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- Performs fairly poorly even though best results are reported

Baseline 2

- GMM-EM
 - run GMM z times, each time with a different parameter initialization, to produce z clusterings
 - set z to 5
 - use EM for parameter estimation

Baseline Results: GMM-EM

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GMM-EM	50.3	67.4	50.7	52.1	69.5	51.8	52.8	55.1	57.1

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- outperforms meta clustering for 3 dimensions
- underperforms meta clustering for 2 dimensions
- does not yield good results overall

Baseline 3

- Davidson & Qi's (2007) algorithm
 - Given a dataset with two clusterings, it assumes one of them is supplied (by a human) as input and aims to produce the other
 - For each dataset with dimensions i and j , we feed it with the gold clustering for dimension i and measure the quality of the clustering it proposes against gold clustering for dimension j
 - Repeat the experiment by reversing the roles of i and j

Baseline Results: Davidson & Qi

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Davidson & Qi	50.9	55.7	50.5	51.2	57.8	50.5	50.9	50.5	51.2

- Does not perform better than the 2 unsupervised baselines

Results: Our PMC Model

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PMC	59.1	61.5	64.8	60.5	63.5	63.0	56.5	64.6	62.2

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- achieves best result for 7 of the 9 dimensions

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PMC	59.1	61.5	64.8	60.5	63.5	63.0	56.5	64.6	62.2

- achieves best result for 7 of the 9 dimensions
- achieves stable results across dimensions

Summary

- presented PMC for producing multiple clusterings
- showed that PMC achieved more stable results along different dimensions of a variety of text datasets than three existing multi-clustering algorithms